ARVIN BOZORGI

Glenelg, MD | (443)-430-5034 | bozorgi@gmail.com | arvinbozorgi.com | GitHub

Education

- *McDonogh High School*, Expected Graduation: 2027
- Dean's List 2024, '25, '26
- 2026 Computer Science Award recipient - Presented to an individual for outstanding achievement and interest in computer science
- 35 ACT

Experience

Glaucoma Detection Research | Research Intern
Mentored by Dr. Mohammad Ali Javidian | June 2025–Present

- Trained PyTorch-based Shifted Window Transformer models to detect glaucoma in retinal images.
- Evaluated five data augmentation techniques to address class imbalance; the best-performing approach reduced false negatives by 33% compared with the baseline.
- Annotated 1,500+ retinal images in Roboflow and trained a model to isolate optic nerve regions.
- Tracked model performance and compared experimental results using Weights & Biases.

UMD Machine Learning Internship, *Intern* | *June 2026 - Present*

- Paid in-person internship at the University of Maryland working under the guidance of Professor Ohadi and his team to predict energy consumption of buildings
- Programmed a script to automate data cleaning for training ML/AI prediction models

UMD Hackathon | *February 2026*

- Competed in a Hackathon hosted by UMD, representing and recognized for my quick problem-solving and my ability to collaborate with a team.

FRC Team, *Programmer for McDonogh School* | *January 2024 - Present*

- Programmer for the school's FRC team, developing robot control software in Java using WPILib and other FRC libraries for competition robots.

Conceptbytes Team, *Developer [Demonstration]* | *August 2024 - Present*

- Collaborate with a well-known influencer and engineer to develop an innovative AI project featured on a professional platform, gaining significant attention.
- Develop computer vision systems using Python, MediaPipe, and OpenCV.

High School STEM Center, *Tutor* | *August 2025 - Present*

- Selected through a competitive interview based on STEM coursework and soft skills to tutor students during school hours.

PUBLICATIONS & PRESENTATIONS

"Impact of Different Data Augmentation Techniques for Glaucoma Detection Using a SWIN Transformer"
First Author | IEEE ICDM 2026, Teen Research Track

- Peer-reviewed paper accepted for publication in the IEEE International Conference on Data Mining (ICDM) 2026 associated proceedings, Teen Research Track.
- Accepted to present findings in China; presentation scheduled for November 14, 2026.

Projects

Computer Vision Chess Glasses | *March 2025*
- Trained a model to detect chessboard corners and warp live footage for analysis.
- Tracked piece movement using binary thresholding and canny edge detection.
- Streamed video from Meta Raybans and displayed moves in real time with PyGame.

Soccer Tic-tac-toe [Demonstration] | *December 2024*
- Hand-annotated 4,000+ images to train a computer vision model that tracks a soccer ball in real time. Used this to project a PyGame tic-tac-toe board onto a wall and detected ball impact location to register moves.
- Created an alternative (projector + laptop + webcam) to $5,000 commercial systems.

Warfields II Home Owners Association Website [Website] | *June 2025*
- Developed a modern and accessible HOA website based on board member feedback.

Jarvis (Hand-Tracking Projected UI), [Demonstration] | *August 2024 - Present*
- Built a menu system and two arcade apps (Space Invaders, Brick Breaker) for Jarvis, an open-source Iron Man–style projected UI using OpenCV, MediaPipe, and PyGame.
- Sold apps to the project founder for $200; featured on his Instagram page.
- Joined the official GitHub developer team and remain an ongoing contributor.

Leadership

Competitive Programming Club, *President & Founder* | *May 2025 - Present*
- Lead discussions on competition strategy and coordinate teams to represent our school in regional programming contests.

Founder of Machine Learning Club, *President & Founder* | *May 2025 - Present*
- Taught neural network programming and machine learning applications.
- Create presentations and Python code walkthroughs for beginner-level models.

Founder of 3D Design Club | *August 2024 - May 2025*
- Founded and led a CAD and 3D printing club, teaching Fusion 360 and prototyping.

Technical Skills:

- **Programming:** Python, Java, C, C#, C++, HTML, JavaScript, CSS, Lua
- **AI & Machine Learning:** TensorFlow, PyTorch, OpenCV, MediaPipe, Weights & Biases, Roboflow, Neural networks, facial recognition, hand tracking
- **3D Design & Prototyping:** Fusion 360, Blender modeling, 3D Printing
- **Certificates:** PCEP – Certified Entry-Level Python Programmer, HarvardX CS50